

SFWRENG 3O03 2023 Midterm Solutions

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Problem 1

We know that for a discrete system that $H(\omega) = e^{-i\omega} - e^{-2i\omega}$. What is the corresponding difference equation for this system?

Solution:

We can use the fact $y(n) = H(\omega)e^{in\omega}$ when $x(n) = e^{in\omega}$ to recover the system's difference equation

$$\begin{aligned} H(\omega) &= e^{-i\omega} - e^{-2i\omega} \\ H(\omega)e^{in\omega} &= e^{in\omega} \cdot (e^{-i\omega} - e^{-2i\omega}) \\ y(n) &= e^{-i\omega} \cdot e^{in\omega} - e^{-2i\omega} \cdot e^{in\omega} \\ &= e^{i\omega(n-1)} - e^{i\omega(n-2)} \\ &= x(n-1) - x(n-2) \end{aligned}$$

Problem 2

What are the Fourier series coefficients of the signal $x(t) = 2 + \sin(t) + \cos(3t) + 2\cos(5t)$? You do not need to compute integrals to get full marks.

Solution:

We will rely on the fact that each term which makes up the signal is periodic, we can use the coefficients of the complex exponential expansions of sine and cosine ($\sin(t) = \frac{e^{it} - e^{-it}}{2i}$, $\cos(t) = \frac{e^{it} + e^{-it}}{2}$) respectively) to find the Fourier series coefficients. Therefore:

$$\begin{aligned}
X_{-5} &= 1 \\
X_{-4} &= 0 \\
X_{-3} &= \frac{1}{2} \\
X_{-2} &= 0 \\
X_{-1} &= \frac{-1}{2i} \\
X_0 &= 2 \\
X_1 &= \frac{i}{2i} \\
X_2 &= 0 \\
X_3 &= \frac{1}{2} \\
X_4 &= 0 \\
X_5 &= 1
\end{aligned}$$

Notice here that:

1. The value of the Fourier series coefficient at index j corresponds to the complex exponential expansion of the term in $x(n)$ with angular frequency j .
2. Angular frequencies which do not appear in the signal are given a value of zero.
3. $X_j = \bar{X}_{-j}$

Problem 3

Define the frequency response of a continuous system.

Solution:

The frequency response, $H(\omega)$ of a continuous system $y(t)$ is defined as $y(t) = H(\omega)e^{it\omega}$ when $x(t) = e^{it\omega}$.

Problem 4

Use convolution to compute the output of a discrete system with impulse response $h(n)$ for the input $x(n) = e^{-in\omega}$. Note that the final result is the *Discrete Time Fourier Series* (DTFT)

Solution:

$$\begin{aligned}
y(n) &= \sum_{k=-\infty}^{\infty} (h(k)x(n-k)) \\
&= \sum_{k=-\infty}^{\infty} (h(k)e^{i\omega(n-k)}) \\
&= \sum_{k=-\infty}^{\infty} (h(k) \cdot e^{-in\omega} \cdot e^{ik\omega}) \implies \\
y(n)e^{in\omega} &= \sum_{k=-\infty}^{\infty} (h(k) \cdot e^{ik\omega}) \implies \\
H(\omega) &= \sum_{k=-\infty}^{\infty} (h(k) \cdot e^{ik\omega})
\end{aligned}$$

Problem 5

Given the difference equation $y(n) = x(n) - y(n-2)$, compute the system's output to the signal $x(n) = 1 + \cos(n\pi + \frac{\pi}{2}) + \sin(n\frac{\pi}{4})$ using the its frequency response.

Solution:

(I) Find the system's frequency response ($H(\omega)$).

$$y(n) = x(n) - y(n-2)$$

$$H(\omega)e^{in\omega} = e^{in\omega} - H(\omega)e^{i\omega(n-2)}$$

$$H(\omega)e^{in\omega} = e^{in\omega} - H(\omega)e^{in\omega}e^{-2i\omega}$$

$$H(\omega) = 1 - H(\omega)e^{-2i\omega}$$

$$H(\omega) + H(\omega)e^{-2i\omega} = 1$$

$$H(\omega)(1 + e^{-2i\omega}) = 1$$

$$H(\omega) = \frac{1}{1 + e^{-2i\omega}}$$

(II) Compute $H(\omega)$, $|H(\omega)|$, and $\arg(H(\omega))$ for each ω in the signal $x(n)$.

ω	$H(\omega)$	$ H(\omega) $	$\arg(H(\omega))$
0	$1/2$	$1/2$	0
π	$1/2$	$1/2$	0
$\pi/4$	$1/(1-i)$	$\sqrt{2}/2$	$-7\pi/4$

(III) Use the formula $y(n) = |H(\omega)| \cos(n\omega + \arg(H(\omega)))$ for each of the terms we computed.

$$\begin{aligned}
 y(n) &= |H(\omega_1)| \cos(n\omega_1 + \arg(H(\omega_1))) + |H(\omega_2)| \cos(n\omega_2 + \arg(H(\omega_2))) \\
 &\quad + |H(\omega_3)| \cos(n\omega_3 + \arg(H(\omega_3))) \\
 &= \frac{1}{2} \cos(0n + 0) + \frac{1}{2} \cos(n\pi + 0) + \frac{\sqrt{2}}{2} \cos\left(\frac{\pi}{4} + \frac{-7\pi}{4}\right) \\
 &= \frac{1}{2} + \frac{1}{2} \cos(\pi) + \frac{\sqrt{2}}{2} \cos\left(\frac{\pi}{4} + \frac{-7\pi}{4}\right)
 \end{aligned}$$

Problem 6

Compute the impulse response and the frequency response of $y(n) = x(n-3)$.

(I) Compute the impulse response ($h(n)$).

Substituting $x(n) = \delta(n)$, we find:

$$\begin{aligned}
 y(n) &= x(n-3) \implies \\
 h(n) &= \delta(n-3)
 \end{aligned}$$

(II) Compute the frequency response ($H(\omega)$).

Using $y(n) = H(\omega)e^{in\omega}$ when $x(n) = e^{in\omega}$, we find:

$$\begin{aligned}
 y(n) &= x(n-3) \\
 H(\omega)e^{in\omega} &= e^{i\omega(n-3)} \\
 H(\omega)e^{in\omega} &= e^{in\omega} \cdot e^{-3i\omega} \\
 H(\omega) &= e^{-3i\omega}
 \end{aligned}$$

Problem 7

What is the output $y(n)$ given the system $y(n) = x(n) + \alpha y(n-1)$ and input $x(n) = 1 \forall n \in \mathbb{N}$?

Solution:

$$y(0) = 1 + \alpha(0) = 1$$

$$y(1) = 1 + \alpha(1) = 1 + \alpha$$

$$y(2) = 1 + \alpha(1 + \alpha) = 1 + \alpha + \alpha^2$$

$$y(3) = 1 + \alpha(1 + \alpha + \alpha^2) = 1 + \alpha + \alpha^2 + \alpha^3$$

Extrapolating, we can surmise that:

$$y(n) = 1 + \alpha + \alpha^2 + \dots + \alpha^n$$

Problem 8

The Fourier series for the 4-periodic signal $x(n) = \sin(\frac{\pi}{2}n)$ was computed and yielded:

$$X_0 = 0, X_1 = -i, X_2 = 0, X_3 = i$$

Compute the values of $x(n)$ for one period.

Solution:

(I) Compute ω_0 .

Since we are given that $x(n)$ is 4-periodic, we can use the formula $\omega_0 = \frac{2\pi}{P}$ to find that:

$$\begin{aligned}\omega_0 &= \frac{2\pi}{4} \\ &= \frac{\pi}{2}\end{aligned}$$

(II) Compute the values of $x(n)$ for one period.

Using the formula:

$$x(n) = \sum_{k=0}^{p=1} (X_k e^{ink\omega_0})$$

we can compute the values of $x(n)$ as follows:

$$x(0) = \sum_{k=0}^3 (X_k e^{i(0)k\frac{\pi}{2}}) = (0 - i + 0 - i) = 0$$

$$x(1) = \sum_{k=0}^3 (X_k e^{i(1)k\frac{\pi}{2}}) = (-ie^{\frac{\pi}{2}i} + ie^{\frac{3\pi}{2}i}) = 2$$

$$x(2) = \sum_{k=0}^3 (X_k e^{i(2)k\frac{\pi}{2}}) = (-ie^{i\pi} + ie^{3i\pi}) = 0$$

$$x(3) = \sum_{k=0}^3 (X_k e^{i(3)k\frac{\pi}{2}}) = (-ie^{\frac{3\pi}{2}i} + ie^{\frac{9\pi}{2}i}) = -2$$